

Image Generation Models

Project: Verified Study Buddy

1. Image Generation Models

Teach the learning cycle

Primary target: Explain the mechanism and terminology of image generation and provenance

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-3/10, ai-3/11. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Sampling and Variability

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

Look for: Sample from a distribution using explicit cumulative intervals

Reference example, not the only acceptable wording: Teacher explanation: Selections red,blue,green,green. The draw 1 violates the half-open $[0,1)$ input contract and must be rejected.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic evidence. Explain what the image can and cannot establish.

Concept to explain

Image generation iteratively constructs pixels guided by learned representations.

Term · Meaning

Term: latent representation

Meaning: An internal representation learned or constructed by a model rather than directly supplied as the original observation.

Term: noise

Meaning: Variation or interference that obscures the information relevant to a task.

Term: denoising

Meaning: Reducing noise in a representation; some generative models learn repeated denoising steps to create outputs.

Term: conditioning

Meaning: Making a model's output depend on supplied information, such as a prompt or image.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can describe a toy denoising process and distinguish generated illustrations from observations.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. In this simplified diffusion-style image generator, begin with random visual noise. Other image-generation methods use different procedures.

2. At each step, predict a slightly more structured image guided by the prompt.

3. Learned patterns influence shapes, textures, and relationships.

4. The result is generated, not retrieved evidence of a real event.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

Exact explanation to model

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems denoise iteratively, but architectures vary.

Explain the added worked example

An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery. Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term · Meaning

Term: Provenance

Meaning: Recorded origin and transformation history

Term: Visual criterion

Meaning: A stated requirement checked against an image

Term: Authenticity claim

Meaning: A claim about the source or reality of depicted content

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs,

evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Image Generation Models

A fictional lesson poster illustrates one family of image-generation approaches.

The facts we will use

Toy 2×2 noisy numerical grid: $[[0.2, 0.8], [0.6, 0.3]]$. A specified illustrative denoising step gives $[[0.1, 0.9], [0.4, 0.2]]$.

These supplied arrays illustrate steps only; they are not outputs from a real trained diffusion model.

Illustrative grids, not executed model outputs

	Left	Right
Before top	0.2	0.8
Before bottom	0.6	0.3
After top	0.1	0.9
After bottom	0.4	0.2

Read a cell using both its row and column labels.

Image Generation Models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Noisy representation

Four toy numerical cells form the starting grid.



2. Condition on request

A text request guides the imagined learned denoising process.



3. Illustrative update

Use the supplied second grid to show one change in representation.



4. Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

CHANGE ONE THING

Relabel the illustration as a photograph of an actual laboratory measurement.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Noisy representation

What happens and why: Four toy numerical cells form the starting grid.

Step: 2. Condition on request

What happens and why: A text request guides the imagined learned denoising process.

Step: 3. Illustrative update

What happens and why: Use the supplied second grid to show one change in representation.

Step: 4. Check provenance

What happens and why: Caption it as a generated-process illustration, not a photographed experiment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Diffusion-style generators learn to remove noise in stages, often guided by text, but a real implementation can operate in a latent representation rather than direct pixels. The toy values let us inspect an update without pretending to simulate that trained process. An attractive final image still needs checks for labels, requested relations and factual claims.

What this example does not establish

Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Change one thing in the pictured case

Change: Relabel the illustration as a photograph of an actual laboratory measurement.

Prediction: That caption is unsupported.

Reason: Its provenance is an invented diagram of a process, not a camera record of an event.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A generator creates a realistic fictional bicycle beside a tree.

Write the question to explain: How does this differ from event photography?

Reveal after predictions: It synthesises an image from learned patterns and instructions; it does not document that event. Many systems denoise iteratively, but architectures vary.

Keep for an independent attempt: An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use. **Qualification:** Provenance tracks production history; it does not prove the content is accurate or establish every legal permission. **Exit:** accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Image generation and provenance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What may these two grids explain, and what can they not establish about a real generator?

Reasoned teacher answer

They explain an illustrative staged change. They cannot establish learned weights, actual image quality or accurate scientific observation.

Changed-case reasoning: That caption is unsupported. Its provenance is an invented diagram of a process, not a camera record of an event.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Order noise-to-image cards, vary one prompt condition, compare composition changes, and label what cannot be inferred as fact.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems denoise iteratively, but architectures vary.

Independent question

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

Expected independent answer

Generated images can miss exact counting constraints. Use a checked, editable drawing or diagram tool for an instructional diagram requiring exactly four roots.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “An image generator always obeys numerical constraints precisely.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

It demonstrates a generated depiction, not that a greenhouse exists on Mars. Label it as generated and separate design imagination from factual evidence. Some image models use iterative denoising, but no single architecture describes every generator.

Explanations for the additional workbook examples

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred. Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery. Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

W2 Trace the worked case

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

W3 Audit the limit

A report omits this qualification: Provenance tracks production history; it does not prove the content is accurate or establish every legal permission. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Provenance tracks production history; it does not prove the content is accurate or establish every legal permission. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

A generated diagram requires two arrows and labels input/output. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental observation. Audit.

Teacher explanation: Arrow count passes; label semantics fail. Metadata supports generation provenance, not observed-event authenticity; replace the caption and correct labels.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. Provenance tracks production history; it does not prove the content is accurate or establish every legal permission. The new case should test this particular mechanism rather than merely restate the original result.

Assess independent transfer

Selected case: A generated diagram requires two arrows and labels input/output. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental observation. Audit.

Selected case answer

Teacher explanation: Arrow count passes; label semantics fail. Metadata supports generation provenance, not observed-event authenticity; replace the caption and correct labels.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

Treat generated diagrams as illustrations; pair factual claims with approved sources.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.
Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 11: Visual Composition. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Arrow count passes; label semantics fail. Metadata supports generation provenance, not observed-event authenticity; replace the caption and correct labels. Require intermediate reasoning and the stated scope. Fresh retry: A generated map has three required labels but omits one and is captioned survey evidence. Expected reasoning: Completeness fails; generation cannot establish surveyed geography. Use verified map data for geographical claims. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Speech pipelines and multimodal evidence. Preserve the current case record and retrieve its evidence boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/11, ai-3/13, ai-3/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Use Study Buddy generated diagrams only with clear provenance and checked educational labels.